Wenhua Wei

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Education

University College London 09/2024 - 09/2025

- MSc Artificial Intelligence for Biomedicine and Healthcare
- Distinction

University College London 09/2018 - 07/2023

- BSc Computer Science
- Upper Second-Class Honours

Research Interests

My research interests center on multimodal understanding, natural language processing, and computer-assisted intervention. I aim to develop scalable and trustworthy clinical agents that perceive, act, and learn continuously and sample-efficiently across surgical and broader clinical settings.

Publications

[1] **Wenhua Wei**, Laurent Mennillo, Zhehua Mao, Anjana Wijekoon, Kendall Feeny, Danyal Zaman Khan, Evangelos B. Mazomenos, Danail Stoyanov, Hani J. Marcus, and Sophia Bano. "MID-POSE: Multi-Instrument Detection and Pose Estimation in Endoscopic Surgery." In *Medical Imaging with Deep Learning (MIDL)*, 2026. **(Accepted)** [PDF]

[2] Kendall Feeny, Anjana Wijekoon, **Wenhua Wei**, Danyal Zaman Khan, Danail Stoyanov, Hani J. Marcus, and Sophia Bano. "A Benchmark of Methods for Surgical Instrument Segmentation in Endoscopic Pituitary Surgery." In *17th International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)*, 2026. **(Accepted)**

Academic Experience

Research Assistant 09/2025 – 12/2025
UCL Hawkes Institute London, UK

- Conducted research on machine perception for surgical instruments in minimally invasive surgery.
- Collaborated with clinicians to construct endoscopic pituitary surgery datasets for multi-class instrument detection, pose estimation, and segmentation.

Postgraduate Teaching Assistant 10/2025 – 12/2025
COMP0245 Machine Learning for Robotics Department of Computer Science, UCL

- Supported lab sessions in applying machine-learning methods to model, control, and optimise robot dynamics using the Franka Emika Panda.
- Graded lab reports and provided feedback.

Research Projects

Multi-Instrument Detection and Pose Estimation in Endoscopic Surgery
Supervisors: Dr. Laurent Mennillo and Dr. Sophia Bano 05/2025 – 12/2025

- Built PitSurg, a monocular endoscopic pituitary surgery dataset with instrument classes, bounding boxes, and 2D keypoints.
- Developed MID-POSE, a single-forward detection and pose estimation model that preserves high-resolution representations, evaluated on PitSurg and the SurgPose robotic surgery dataset to assess cross-dataset generalisation.

- Outperformed YOLOv8x-Pose, the strongest baseline for this task, in $\text{mAP}_{50\text{–}95}$ on PitSurg and SurgPose, improving detection/pose from 59.4%/63.1% to 77.5%/78.5% and from 47.9%/61.1% to 62.7%/71.4%.

Quantitative Water-Fat Separation from Multi-Echo MRI Magnitude Data
Supervisor: Dr Timothy Bray 03/2025 – 04/2025

- Built a magnitude-only multi-echo MRI estimator for water–fat separation and PDFF mapping using constrained least-squares estimation of voxelwise water and fat amplitudes and R_2^*. Diagnosed water-fat swaps driven by dual optima.
- Upgraded the estimator to Rician maximum-likelihood fitting, reducing water–fat swap frequency by ~62%, and yielding smoother PDFF and R_2^* maps in low-signal and fast-decay regions.

Brain Microstructure Estimation via Ball-and-Stick Modelling of Diffusion MRI (dMRI)
Supervisor: Prof. Daniel Alexander 11/2022 – 05/2023

- Developed a self-supervised multilayer perceptron (MLP) to estimate brain microstructure parameters using the ball-and-stick diffusion model from dMRI signals.
- MLP achieved $29\times$ faster inference than conventional nonlinear least-squares fitting and produced smoother parameter maps. With 50% fewer measurements, it remained close to the full-measurement reference for volume fraction and diffusivities with SSIM $\geq 98.8\%$ and MSE $\leq 5 \times 10^{-4}$. Stick orientation reached $\sim 92\%$ SSIM and ~ 0.93 cosine similarity, demonstrating the clinical promise of learning-based microstructure estimation by reducing the required measurements and therefore scan time.

Industry Experience

AI Engineer Intern 09/2023 – 08/2024
AI4C Applied Research Institute Guangzhou, China

- Led development of an Alzheimer's cognitive-care conversational agent with proactive dialogue and gradual personalisation using Microsoft Semantic Kernel and the OpenAI API. Registered with the National Copyright Administration of China and piloted in 10 nursing homes.
- Delivered an internal and client-facing AI consulting agent supporting MVP planning and structured requirements generation.

Software Engineer Intern 06/2020 – 09/2020
Huawei Technologies Co., Ltd, Department of Public Development Dongguan, China

- Improved observability for a Kafka messaging platform by adding 7 production metrics and building 9 ZooKeeper health-check scripts for automated diagnostics.
- Designed and implemented automatic endpoint updates on IP failure using Apache Curator, removing manual IP changes and enabling microsecond-level switchover.

Software Engineer Intern 06/2019 – 08/2019
Institute of Automation, Chinese Academy of Sciences (CASIA) Beijing, China

- Built a Raspberry Pi autonomous lane-following model car for the Smart Education Research Group and taught it at CASIA Summer Camp.

Skills

Programming Languages
Python, C#, MATLAB, Java, C, Haskell

Frameworks & Libraries
PyTorch, scikit-learn, Hugging Face, OpenCV, NumPy, Pandas

Robotics and 3D Vision
ROS 2, ORB-SLAM2, COLMAP

Operating Systems
Linux, Windows, macOS

Cloud & DevOps
Microsoft Azure, Docker, Git, GitHub